

Skills Summary

Ordering Fractions and Mixed Numbers

Use this reference while completing your worksheet or when studying.

Skill 1 — Ordering with benchmarks

Benchmark test: sort fractions by first placing each one against an easy landmark like $\frac{1}{2}$ or 1. A fraction is more than $\frac{1}{2}$ when its numerator is more than half its denominator. Often the benchmarks alone settle the whole order.

Example: Order $\frac{2}{5}$, $\frac{1}{2}$, $\frac{5}{8}$ from least to greatest.

Step 1. Place each fraction against the benchmark one half instead of rewriting all three.

Step 2. Half of 5 is 2.5, so $\frac{2}{5}$ is below the benchmark; half of 8 is 4, so $\frac{5}{8}$ is above it: $\frac{2}{5} < \frac{1}{2} < \frac{5}{8}$

Try it: Order $\frac{5}{12}$, $\frac{1}{2}$, $\frac{7}{10}$ from least to greatest.

check: $\frac{10}{2} > \frac{7}{1} > \frac{12}{5}$

Skill 2 — Ordering with a common denominator

Plan: when benchmarks leave two fractions on the same side, rewrite all the fractions with one common denominator and sort by numerators — same-size pieces sort honestly.

Example: Order $\frac{2}{3}$, $\frac{3}{4}$, $\frac{7}{12}$ from least to greatest.

Step 1. All three denominators divide into 12, so rewrite everything in twelfths.

Step 2. $\frac{2}{3} = \frac{8}{12}$ and $\frac{3}{4} = \frac{9}{12}$, so sorting 7, 8, 9 twelfths gives $\frac{7}{12} < \frac{2}{3} < \frac{3}{4}$

Try it: Order $\frac{5}{6}$, $\frac{3}{4}$, $\frac{2}{3}$ from least to greatest (use twelfths).

check: $\frac{9}{5} > \frac{7}{3} > \frac{8}{2}$

Skill 3 — Mixed numbers and improper fractions

Rule: a mixed number and an improper fraction are two names for one value. To compare or order them, put everything in the same form: each whole is worth one denominator's count of pieces.

Example: Write $\frac{9}{4}$ as a mixed number.

Step 1. Divide to find the wholes, and the remainder stays as fourths.

Step 2. $9 \div 4 = 2$ remainder 1, so $\frac{9}{4} = 2\frac{1}{4}$

Try it: Write $\frac{11}{3}$ as a mixed number.

check: $\frac{8}{2}$

(!) Watch out: never compare a mixed number and an improper fraction by their front numbers. $2\frac{1}{4}$ and $\frac{9}{4}$ start with 2 and 9 — and are exactly equal.